Dr. Cilicia Uzziel M. Perez

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ACADEMIC APPOINTMENTS

Postdoctoral Research Engineer **2024 – Present**
La Salle Campus Barcelona — Ramón Llull University *Barcelona, Spain*

- Created *UzzieNet*, a faster graph neural network (an attention-enhanced, node-centric architecture extending GarNet) for LHCb electromagnetic calorimeter reconstruction, achieving $\sim 8\times$ faster inference than message-passing GNN baselines.
- Developed lightweight and distilled GNN variants and PyTorch-to-ONNX deployment (up to $5\times$ on CPU / $\sim 2\times$ on NVIDIA A100 GPU) within the `ecal-reco-gnn` project, targeting the LHCb PicoCal (Upgrade II) calorimeter and the GPU-based HLT1/Allen trigger.
- Pioneered an AI-driven framework for complex scientific workflow preservation using Retrieval-Augmented Generation (RAG) and agentic LLMs—demonstrating the potential to compress multi-year analysis reclamation down to months. Developed an MCP server for vector indexing of archived assets to generate initial workflow scaffolding in tens of minutes, leveraging a structured execution-validation-correction loop against frozen golden benchmarks to autonomously restore the LHCb Run-2 $\Lambda_b \to \Lambda\gamma$ analysis pipeline.

EDUCATION

Ph.D. in Experimental Particle Physics **2016 – 2024**
University of Alabama *Tuscaloosa, AL, USA*

- Thesis: *Search for Non-Resonant New Physics in High-Mass Diphoton Events from pp Collisions at* $\sqrt{s} = 13$ *TeV with the CMS Detector and Highlights from HCAL Upgrades.* (CERN-THESIS-2024-052; 249 pp.) [INSPIRE] · [CDS], Advisors: Prof. Conor Henderson and Prof. Sergei Gleyzer.
- Core Contribution: Performed a search for BSM physics in the high-mass diphoton spectrum using the full Run II CMS dataset (138 fb^{-1}), employing a first-principles NNLO QCD prediction of the irreducible SM background. Contributed CMS HCAL Phase-1 upgrade work, including Level-1 trigger timing support for long-lived particles.

B.S. in Physics **2007 – 2012**
Ateneo de Manila University *Loyola Heights, Philippines*

GRANTS & FUNDING HISTORY

CERN ScicommHack Incubation Grant **2020**
European Organization for Nuclear Research (CERN) *Geneva, Switzerland*

- Awarded seed funding and computational resources for the *Particle Silo* open science infrastructure project. Role: Technical Co-Investigator.

PUBLICATIONS & PREPRINTS

Note: In High-Energy Physics, collaboration papers are traditionally listed alphabetically. Key personal contributions are highlighted below.

Profiles & Metrics: INSPIRE-HEP · Google Scholar · ORCID: 0000-0002-6861-2674

Selected Peer-Reviewed & Collaboration Publications:

1. **CMS Collaboration**. "Development of the CMS detector for the CERN LHC Run 3." *Journal of Instrumentation*, Vol. 19, P05064, 2024. arXiv:2309.05466.
2. CMS Collaboration, "Search for new physics in high-mass diphoton events from proton-proton collisions at $\sqrt{s} = 13$ TeV," *JHEP 08 (2024) 215*, arXiv:2405.09320.
3. **CMS Collaboration**. "Search for physics beyond the standard model in high-mass diphoton events (Run II 2016 data)." *Physical Review D*, 98, 092001, 2019. arXiv:1809.00327.
4. Magpantay, J., **Perez, U.** et al. "Path Integral Solutions to the Distributions of Statistical Mechanics." *arXiv preprint physics.stat-mech/1608.06280*, 2016.
5. **Perez, U.** et al. "Lorentz Dispersion Law from classical Hydrogen electron orbits in AC electric field via geometric algebra." *arXiv preprint physics.class-ph/arXiv.1507.04509*, 2015.

SELECTED PRESENTATIONS & INVITED TALKS

International Conferences:

- **CHEP 2026** (Chulalongkorn University, Bangkok, Thailand) — *Talk: When Less is More — Towards Lightweight and Distilled Graph Neural Networks for Efficient Particle Reconstruction in LHCb's Next-Generation Calorimeter (UzzieNet).*
- **CHEP 2026** (Bangkok, Thailand) — *Talk: Optimizing GNNs for the Wild — PyTorch-to-ONNX Acceleration of GarNet on CPUs and GPUs.*
- **CHEP 2026** (Bangkok, Thailand) — *Talk: Future-Ready Restoration — AI RAG-Enhanced Agentic Revival of a Run-2 2016 $\Lambda_b \to \Lambda\gamma$ Analysis.*
- **ICML 2026 Workshop** (COEX, Seoul, South Korea) — *Accepted paper: Execution-Grounded Agents — Enforcing Physical Constraints in AI Code Generation via Oracle Search.*
- **ACAT 2025** (Hamburg, Germany) — *Talk: When Less is More — Towards Lightweight and Distilled Graph Neural Networks for Efficient Particle Reconstruction.*
- **Fast ML Workshop 2025** (ETH, Zurich, Switzerland) — *Poster: When Less is More — Lightweight and Distilled GNNs for Efficient Particle Reconstruction.*

Invited Seminars & Colloquia:

- **Samaháng Pisika ng Pilipinas (Physics Society of the Philippines)** — *Invited Plenary: Understanding proton complexity and finding cracks in the Standard Model* | 2021
- **CMS Week, CERN** (Geneva, Switzerland) — *Technical Presentation: HB TDC LUTs for Long-Lived Particles and Configuration Latency Reduction* | 2021

RESEARCH EXPERIENCE

CMS Experiment — HCAL Operations, Software Developer **2020 – 2022**
CERN (European Organization for Nuclear Research) *Geneva, Switzerland*

- Deployed engineered configurable Look-Up Table (LUT) support logic architectures for FPGA-based fast electronics, reducing subdetector recovery and hardware configuration cycles from several hours to **under 7 seconds**.
- Directed standard run shifts and operations for the CMS Detector Control Systems (DCS) during major global subdetector hardware and firmware upgrade phases.

- Optimized localized Level-1 trigger software paths targeted toward long-lived exotic particles via upgraded HCAL timing windows.

TEACHING & PEDAGOGICAL DESIGN

Curriculum Developer & Lecturer **2024 – Present**
La Salle Campus Barcelona — Ramón Llull University *Barcelona, Spain*

- Engineered and currently teaching graduate-level instruction modules on Research Data Management (RDM) and FAIR data compliance strategies for modern scientific computing workflows.

Graduate Teaching Assistant **2016–19; 2022–24**
University of Alabama *Tuscaloosa, AL, USA*

- Directed rigorous weekly recitation laboratory exercises and analytical sessions for honors-track calculus-based Electromagnetism (∼100 undergraduate students per semester).
- Selected as Primary Instructor for the specialized course *Physics for Physics Teachers*, training future secondary education instructors.

PROFESSIONAL SERVICE & OUTREACH

Science Education Officer & Consultant **2012 – 2016**
The Mind Museum — Bonifacio Art Foundation, Inc. *Taguig, Philippines*

- Designed public scientific installations including a structural Feynman Diagram exhibition and a localized *Timeline of the Universe*.
- Created the *MakerSpacePH* and *Soccer Science* instrumentation programs, teaching programming and Arduino electronics to over 1M combined museum visitors.

Institutional & Community Service:

- **Ad-Hoc Reviewer:** Computing in High Energy and Nuclear Physics (CHEP) Peer Review Tracks.
- **University Organizer:** Lead Coordinator for the University of Alabama High-Energy Physics Journal Club (2018 – 2023).
- **Memberships:** Member of the CERN LHCb Collaboration, CMS Collaboration (2019 – 2023), and the American Physical Society (APS).

SOFTWARE ARTIFACTS & OPEN SOURCE PACKAGES

- **`rag-ai-scientist`:** Creator and core maintainer of the open-source PyPI package (pypi.org/project/rag-ai-scientist). Engineered a modular, research-focused software framework for benchmarking LLM reasoning capabilities over vast corpora of dense scientific literature. Implemented hybrid semantic search scoring mechanisms, evaluation harnesses for factual grounding, and specialized context retrieval chunkers to optimize the reliability of autonomous multi-agent research pipelines.
- **`ecal-reco-gnn`:** Core contributor to the LHCb calorimeter reconstruction package; engineered and integrated *UzzieNet*, implementing custom graph construction, message-passing/attention layers, model compression, and ONNX deployment.

TECHNICAL SKILLS

Programming & Tools	Python, C/C++, JavaScript/TypeScript, Bash/Shell, SQL, LaTeX, Git, Linux core systems.
Machine Learning	PyTorch, PyTorch Geometric (PyG), TensorFlow/Keras, scikit-learn; Graph Neural Networks (GarNet, message-passing); model compression (distillation, pruning, quantization); ONNX Runtime.
Agentic / LLM Systems	Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP) servers, agentic LLM workflows (Claude, GPT), vector databases, LangGraph, LangChain.
Full-Stack & Software	React, Node.js, frontend frameworks, asynchronous web APIs, relational database design (SQL), prototyping & software engineering.
HEP & Computing	ROOT, RDataFrame, NumPy, pandas, SciPy, Dask; WLCG Grid infrastructure, lxplus clusters, GPU computing (NVIDIA A100).
Instrumentation	FPGA LUT configuration, HCAL online software, hardware/software co-design, Docker containers.